

Chapter 03 Homework — Rings

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Problem 03.1. Suppose that S is a set and R is a ring. Let R^S denote the set of all mappings from S to R . Show that R^S is a ring, under the operations defined by

$$(f + g)(s) = f(s) + g(s), \quad (fg)(s) = f(s)g(s).$$

Show that if S has more than one element then there exist non-zero elements f and g in R^S for which $fg = 0$.

Solution. Write $f_1, f_0 \in R^S$ for the functions

$$f_1(s) = 1_R, \quad f_0(s) = 0_R,$$

and clearly, by the specified function properties, we have all ring properties.

Prove $(f + g)h = fh + gh$. Let $s \in S$. Show $((f + g)h)(s) = (fh)(s) + (gh)(s)$:

$$((f + g)h)(s) = (f(s) + g(s))h(s) = f(s)h(s) + g(s)h(s) = (fh)(s) + (gh)(s).$$

If $|S| > 1$, there exist $s_1, s_2 \in S$ where $s_1 \neq s_2$. Let

$$f(s) = \begin{cases} 1_R & s = s_1 \\ 0_R & \text{else} \end{cases} \quad g(s) = \begin{cases} 1_R & s = s_2 \\ 0_R & \text{else} \end{cases}$$

Then $f(s)g(s) = 0$, so they are zero divisors.

Problem 03.2. Suppose that R is an integral domain, with field of fractions F . Show that the field of fractions of $R[x_1, \dots, x_n]$ can be identified naturally with $F(x_1, \dots, x_n)$.

Problem 03.3. Show that an integral domain with a finite number of elements is always a field.

Solution. Let R be a finite integral domain and let $x \in R$ be non-zero.

Consider the sequence of powers $x, x^2, x^3, \dots$. Every term is non-zero: $x \neq 0$, and an integral domain has no zero divisors, so $x^n \neq 0$ for all n . The powers therefore all lie among the non-zero elements of R , of which there are only finitely many, so the sequence must repeat.

Say the repeat is $x^k = x^\ell$ with $\ell > k$. Writing the right-hand side as $x^k x^{\ell-k}$ gives

$$x^k x^{\ell-k} = x^k,$$

and cancelling x^k — which is legitimate, x^k being non-zero in an integral domain — leaves

$$x^{\ell-k} = e.$$

Hence $x \cdot x^{\ell-k} = x$, so the sequence does loop back to x : there is an n , namely $n = \ell - k + 1$, with $x^n = x$.

Now

$$x^n = x \iff x^{n-1}x = x \iff x^{n-1} = e \iff x^{n-2}x = e,$$

the middle step again being cancellation of x . So x^{n-2} is the inverse of x , and it is a genuine power since $\ell > k$ forces $n \geq 2$.

Every non-zero element of R therefore has an inverse, and R is a field.

Problem 03.8. Suppose that K is a field. If $f = a_0 + a_1x + \cdots + a_nx^n \in K[x]$ and $k \in K$, let $(\Phi(f))(k) = a_0 + a_1k + \cdots + a_nk^n$. Show that Φ is a ring homomorphism from $K[x]$ to K^K . Show that if K is finite then Φ is an epimorphism, but not a monomorphism. What happens if K is infinite?

Solution. Let $f = \sum_i a_i x^i$ and $g = \sum_i b_i x^i$ in $K[x]$.

Additive half. Let $k \in K$. Then

$$\begin{aligned} (\Phi(f+g))(k) &= \left(\Phi\left(\sum_i (a_i + b_i)x^i\right) \right)(k) = \sum_i (a_i + b_i)k^i = \sum_i a_i k^i + \sum_i b_i k^i \\ &= \left(\Phi\left(\sum_i a_i x^i\right) \right)(k) + \left(\Phi\left(\sum_i b_i x^i\right) \right)(k) = (\Phi(f))(k) + (\Phi(g))(k). \end{aligned}$$

As $k \in K$ was arbitrary, $\Phi(f+g) = \Phi(f) + \Phi(g)$ in K^K .

Multiplicative half. Let $k \in K$. Then

$$\begin{aligned} (\Phi(fg))(k) &= \left(\Phi\left(\sum_m \left(\sum_{i+j=m} a_i b_j\right) x^m\right) \right)(k) = \sum_m \left(\sum_{i+j=m} a_i b_j\right) k^m \\ &= \left(\sum_i a_i k^i\right) \left(\sum_j b_j k^j\right) = \left(\Phi\left(\sum_i a_i x^i\right)\right)(k) \cdot \left(\Phi\left(\sum_j b_j x^j\right)\right)(k) = (\Phi(f))(k) (\Phi(g))(k). \end{aligned}$$

Φ is an epimorphism when K is finite. Let K be a finite field and let $h \in K^K$ be arbitrary. We find $f \in K[x]$ with $(\Phi(f))(k) = h(k)$. Take

$$f(x) = \sum_k h(k) \delta_k(x).$$

Let $k_1 \in K$. Then

$$(\Phi(f))(k_1) = \left(\Phi\left(\sum_k h(k) \delta_k(x)\right) \right)(k_1) = \sum_k h(k) \delta_k(k_1) = h(k_1) \delta_{k_1}(k_1) = h(k_1) \cdot 1 = h(k_1).$$

So Φ is an epimorphism.

Φ is not a monomorphism. Take $K = \mathbb{F}_2$, $f = x^2 + x$ and $g = 0$. Then $f \neq g$ in $\mathbb{F}_2[x]$, but

$$(\Phi(f))(0) = 0 = (\Phi(g))(0), \quad (\Phi(f))(1) = 1 + 1 = 0 = (\Phi(g))(1),$$

so $\Phi(f) = \Phi(g)$ in $\mathbb{F}_2^{\mathbb{F}_2}$ and $\ker \Phi \neq \{0\}$.

For an arbitrary finite field K , take

$$f(x) = \prod_{k \in K} \delta_k(x),$$

a non-zero element of $\ker \Phi$.

Problem 03.10. What are the units in $R[x]$, where R is an integral domain? What are the units in $\mathbb{Z}_4[x]$?

Solution. The units in $R[x]$, for R an integral domain, are

$$\{a \in R \subseteq R[x] : a^{-1} \text{ exists}\},$$

the existence of a^{-1} being what is not guaranteed in an integral domain.

They must be constants. For in an integral domain R , with $f, g \in R[x]$,

$$\deg(fg) = \deg(f) + \deg(g),$$

since no non-zero coefficients multiply to zero. So $\deg(f) = \deg(g) = 0$.

In $\mathbb{Z}_4[x]$ the units are 1 and 3, since $3 \cdot 3 = 9 \equiv 1$; and also, taking

$$f(x) = 2x + 3, \quad g(x) = 2x + 3,$$

we have $fg = 4x^2 + 12x + 9 = 1$.

So in $\mathbb{Z}_4[x]$ the units are those f whose constant term is a unit and all of whose other coefficients are zero divisors: constant term in $\{1, 3\}$, every other coefficient in $\{0, 2\}$.

Problem 03.11. Show that an element a of an integral domain R is prime if and only if $R/(a)$ is an integral domain.

Solution. ($\Rightarrow$) Suppose $a \in R$ is prime. Show $R/(a)$ is an integral domain.

Consider

$$(r_1 + (a))(r_2 + (a)) = r_1r_2 + (a).$$

Take the factors non-zero, so $a \nmid r_1$ and $a \nmid r_2$ — else we would have zero divisors. Then $r_1r_2 + (a)$ is non-zero, as $a \nmid r_1r_2$ since a is prime. So $R/(a)$ has no zero divisors.

Further, $1 + (a) \neq 0 + (a)$, because $a \nmid 1$ (a is not a unit).

($\Leftarrow$) Suppose $R/(a)$ is an integral domain. Show $a \in R$ is prime.

Suppose a is not prime. So there exist $b, c \in R$ such that $a \mid bc$, but $a \nmid b$ and $a \nmid c$. Note $b + (a)$ and $c + (a)$ lie in $R/(a)$, both non-zero since $a \nmid b$, $a \nmid c$. And

$$(b + (a))(c + (a)) = bc + (a) = (a)$$

since $a \mid bc$. So $R/(a)$ has zero divisors and is not an integral domain.

Further, a is not a unit. Suppose $a \in R$ is a unit; show $R/(a)$ is not an integral domain. Note $a^{-1} \in R$, so $aa^{-1} = e \in (a)$. Let $r \in R$. Note $r \in (a)$ if there exists $x \in R$ such that $xa = r$; take $x = ra^{-1}$, which exists since a is a unit in R . So $(a) = R$, and

$$R/(a) = R/R = \{R\},$$

a single-element set, which cannot be an integral domain since it is a ring with $e = 0$.

Problem 03.12. Let $R = \mathbb{Z} + i\sqrt{5}\mathbb{Z}$.

1. Show that the units are 1 and -1 .

2. Let $\varphi(m + i\sqrt{5}n) = m^2 + 5n^2$. Then $\varphi(0) = 0$, $\varphi(1) = \varphi(-1) = 1$ and otherwise $\varphi(\alpha) > 3$. Use this to show that $2 + i\sqrt{5}$ and $2 - i\sqrt{5}$ are irreducible in R .

3. Show that $2 + i\sqrt{5}$ is not a prime; R is not a unique factorization domain.

Solution. (a) Both 1 and -1 are units, each being its own inverse.

Conversely, suppose $\alpha \in R$ is a unit, so $\alpha\beta = 1$ for some $\beta \in R$. Then

$$\varphi(\alpha\beta) = \varphi(1) = 1,$$

and so $\varphi(\alpha)\varphi(\beta) = 1$. Since $\varphi : R \rightarrow \mathbb{N}$, this gives $\varphi(\alpha) = 1$ and $\varphi(\beta) = \varphi(\alpha)$. Through the definition of φ , this forces $m = \pm 1$, $n = 0$; that is, $\alpha = \pm 1$.

(b) Write $\alpha_1 = 2 + i\sqrt{5}$ and $\alpha_2 = 2 - i\sqrt{5}$.

Neither is a unit, the units being ± 1 by part (a).

Suppose $\alpha_1 = \beta\gamma$ with β and γ both not units. Now

$$\varphi(2 + i\sqrt{5}) = 4 + 5 = 9,$$

and the only ways to write 9 as a product of two elements of $\mathbb{N}$ are $3 \cdot 3$ and $1 \cdot 9$. Since

$$\varphi(\beta\gamma) = \varphi(\beta)\varphi(\gamma) = 9,$$

one of these two splittings must occur.

Could $m^2 + 5n^2 = 3$ for $m, n \in \mathbb{Z}$? No: $n = 0$ leaves $m^2 = 3$, and $|n| \geq 1$ makes the value at least 5. So the splitting $3 \cdot 3$ cannot occur, and therefore $\varphi(\beta) = 1$ or $\varphi(\gamma) = 1$ — which means β or γ is a unit, contrary to assumption.

Hence $2 + i\sqrt{5}$ is irreducible. The same argument applies verbatim to $2 - i\sqrt{5}$, since $\varphi(2 - i\sqrt{5}) = 9$ also.

(c) Write $\alpha = 2 + i\sqrt{5}$. To show α is not prime it is enough to find $\beta, \gamma \in R$ with $\alpha \mid \beta\gamma$ but $\alpha \nmid \beta$ and $\alpha \nmid \gamma$; equivalently, to exhibit some $c_1 \in R$ with $\alpha c_1 = \beta\gamma$ while no $c_2 \in R$ satisfies $\alpha c_2 = \beta$ or $\alpha c_2 = \gamma$.

Take $\beta = \gamma = 3$, so $\beta\gamma = 9$. Then

$$(2 + i\sqrt{5})(2 - i\sqrt{5}) = 9,$$

and $2 - i\sqrt{5} \in R$, so $\alpha \mid \beta\gamma = 9$.

But $\alpha \nmid 3$. Such a divisibility would require some $c_2 \in R$ with $\alpha c_2 = 3$, and hence

$$\varphi(\alpha)\varphi(c_2) = \varphi(3) \iff 9\varphi(c_2) = 9 \implies \varphi(c_2) = 1,$$

which by part (a) forces $c_2 = \pm 1$. Neither works: $(2 + i\sqrt{5})(\pm 1) \neq 3$.

So $\alpha = 2 + i\sqrt{5}$ is not prime.

Now α is an irreducible element of R by part (b), but it is not prime. In a unique factorization domain every irreducible element is prime, so $R = \mathbb{Z} + i\sqrt{5}\mathbb{Z}$ is not a unique factorization domain.

Problem 03.14. A proper ideal I of a ring R is said to be prime if whenever $ab \in I$ then either $a \in I$ or $b \in I$. Show that a non-zero element c of R is prime if and only if (c) is a prime ideal.

Solution. Throughout, $(c) = \{rc : r \in R\}$, and $a \in (c) \iff c \mid a$.

($\implies$) Suppose c is prime. Let $ab \in (c)$. So there exists $r \in R$ with $cr = ab$, that is, $c \mid ab$. Since c is prime, $c \mid a$ or $c \mid b$. Without loss of generality suppose $c \mid a$. Then there exists $r_1 \in R$ with $r_1c = a$, and so by definition $a \in (c)$.

Now we must show that (c) is a proper ideal. Note c is prime, and hence not a unit. So $(c) \neq R$, and (c) is therefore a proper ideal.

($\Leftarrow$) Suppose (c) is a prime ideal. Suppose $c \mid ab$. Then there exists $r \in R$ with $cr = ab$. So $ab \in (c)$, meaning $a \in (c)$ or $b \in (c)$. Without loss of generality suppose $a \in (c)$. Then there exists $r_1 \in R$ with $cr_1 = a$, so $c \mid a$.

To complete this direction, show c is not a unit. Now (c) is a prime ideal by assumption, so it is a proper ideal of R , meaning c can't be a unit.

Problem 03.15. Suppose that $f = k_0 + k_1x + \cdots + k_nx^n$ is a non-zero element of $K[x]$ (where K is a field). An element a of K is a root of f if $f(a) = k_0 + k_1a + \cdots + k_na^n = 0$. Show that a is a root of f if and only if $f \in (x - a)$, and show that f has at most n distinct roots.

Solution. Throughout, K is a field, $a \in K$, and $(x - a)$ denotes the principal ideal $\{g \cdot (x - a) : g \in K[x]\}$ of $K[x]$, so that $f \in (x - a)$ means precisely that $f = g \cdot (x - a)$ for some $g \in K[x]$.

($\Leftarrow$) Suppose $f \in (x - a)$; show $f(a) = 0_K$. There exists $g \in K[x]$ with $g \cdot (x - a) = f$. Evaluating at a , and using that evaluation respects products,

$$f(a) = g(a) \cdot (a - a) = g(a) \cdot 0_K = 0_K,$$

so a is a root of f .

($\Rightarrow$) Suppose $f(a) = 0$; show there exists $q \in K[x]$ with $f = (x - a)q$.

If $f = 0$ then $f = 0 \cdot (x - a)$ and we are done, so suppose $f \neq 0$. Both f and $x - a$ are non-zero elements of $K[x]$, so Theorem 3.7 applies: there exist $q, r \in K[x]$ with

$$f = (x - a)q + r, \quad \text{where } r = 0 \text{ or } \deg r < \deg(x - a) = 1.$$

If $r = 0$ we are done. Otherwise $\deg r < 1$ forces $\deg r = 0$, so r is a non-zero constant.

Now evaluate at a . Since evaluation respects sums and products, and since a constant evaluates to itself,

$$0 = f(a) = (a - a)q(a) + r = 0 \cdot q(a) + r = r,$$

contradicting $r \neq 0$. Hence $r = 0$ after all, and $f = (x - a)q$, that is, $f \in (x - a)$.

At most n distinct roots. By induction on $n = \deg f$, over all non-zero $f \in K[x]$.

Base case, $n = 0$. Then $f = k_0$ with $k_0 \neq 0$, and for every $k \in K$ we have $f(k) = k_0 \neq 0$. So f has no roots at all, and $0 \leq 0 = n$.

Inductive step. Let $n \geq 1$ and suppose every non-zero polynomial in $K[x]$ of degree $n - 1$ has at most $n - 1$ distinct roots. Let f be non-zero of degree n .

If f has no root in K then it has $0 \leq n$ distinct roots and there is nothing to prove. Otherwise pick a root $a \in K$. By the biconditional just proved, $f = (x - a)g$ for some $g \in K[x]$; and $g \neq 0$ since $f \neq 0$. Since K is a field it is an integral domain, so by Exercise 3.10,

$$n = \deg f = \deg(x - a) + \deg g = 1 + \deg g, \quad \text{hence } \deg g = n - 1.$$

Now let $b \in K$ be any root of f with $b \neq a$. Evaluating at b ,

$$0 = f(b) = (b - a)g(b).$$

Here $b - a \neq 0$ because $b \neq a$, and K has no zero divisors, so $g(b) = 0$. Thus every root of f is either a or a root of g .

By the inductive hypothesis g has at most $n - 1$ distinct roots, so the set of roots of f is contained in $\{a\}$ together with a set of at most $n - 1$ elements. Hence f has at most n distinct roots, completing the induction.

Problem 03.17. Suppose that p is a prime number. Let R be the set of rationals which can be written in the form r/s , where p does not divide s . Show that R is a subring of $\mathbb{Q}$. What are the units in R ? Show that R is a principal ideal domain.

Solution. Throughout, p is the fixed prime, and

$$R = \{x \in \mathbb{Q} : x = r/s \text{ for some } r, s \in \mathbb{Z}, s \neq 0, p \nmid s\}.$$

Note that membership in R asks only for *some* representation of x with $p \nmid s$; a single bad representation proves nothing. For instance $6/3$ lies in R for $p = 3$, since $6/3 = 2/1$.

R is a subring of $\mathbb{Q}$. We check the four conditions.

$1 \in R$. Since $p > 1$ we have $p - 1 \neq 0$, so $1 = (p - 1)/(p - 1)$ is a representation of 1 of the required form. Its denominator is not divisible by p : the smallest positive multiple of p is p itself, and $0 < p - 1 < p$, so no multiple of p equals $p - 1$. Hence $1 \in R$.

Closed under addition. Let $x, y \in R$, say $x = a/b$ and $y = c/d$ with $p \nmid b$ and $p \nmid d$. Then

$$x + y = \frac{ad + bc}{bd},$$

and $bd \neq 0$. Since p is prime and $p \nmid b$, $p \nmid d$, Euclid's lemma in contrapositive form gives $p \nmid bd$. So this representation of $x + y$ has denominator not divisible by p , and $x + y \in R$.

Closed under negation. Let $x = a/b \in R$ with $p \nmid b$. Then $-x = (-a)/b$: the sign moves to the numerator and the denominator is unchanged. Membership in R constrains only the denominator, so $-x \in R$.

Closed under multiplication. With $x = a/b$ and $y = c/d$ as above,

$$xy = \frac{ac}{bd},$$

and again $p \nmid bd$ by the contrapositive of Euclid's lemma. So $xy \in R$.

Hence R is a subring of $\mathbb{Q}$. Being a subring of a field, R is a commutative ring with no zero divisors, that is, an integral domain.

The units of R . We claim

$$R^\times = \{n/q \in R : p \nmid n\}.$$

The condition is well defined. Write $x \in R$ in lowest terms as a/b . Every representation of x is $(ra)/(rb)$ for some non-zero integer r , and such a representation is admissible ($p \nmid rb$) exactly when $p \nmid r$ and $p \nmid b$. In that case $p \mid ra$ if and only if $p \mid a$, by Euclid's lemma. So whether p divides the numerator does not depend on which admissible representation of x we chose.

($\supseteq$). Let $x = n/q \in R$ with $p \nmid q$ and $p \nmid n$. Since $p \nmid 0$, the hypothesis $p \nmid n$ forces $n \neq 0$, so q/n is a rational number. Its denominator n satisfies $p \nmid n$, so $q/n \in R$, and

$$\frac{n}{q} \cdot \frac{q}{n} = 1.$$

Hence x is a unit of R .

($\subseteq$). We prove the contrapositive: if $n/q \in R$ and $p \mid n$, then n/q is not a unit. By the previous paragraph we may assume n/q is in lowest terms, so $\gcd(n, q) = 1$.

The inverse of n/q in $\mathbb{Q}$ is q/n , and inverses in a field are unique; so if n/q had an inverse in R , that inverse would have to be q/n , and in particular q/n would lie in R . We show it does not, by ruling out *every* representation of it.

Since $\gcd(q, n) = 1$, the integer fractions equal to q/n are exactly the fractions $(rq)/(rn)$ for non-zero integers r . Now $p \mid n$ means $n = pk$ for some integer k , so for every non-zero integer r ,

$$r \cdot n = p \cdot (rk),$$

which exhibits rk as a witness to $p \mid rn$. So every representation of q/n has denominator divisible by p , and therefore $q/n \notin R$. Hence n/q is not a unit of R .

Combining the two inclusions, the units of R are precisely the elements of R whose numerator, in lowest terms, is not divisible by p .

Problem 03.25. *[TODO: statement not yet transcribed]*

Problem 03.26. *[TODO: statement not yet transcribed]*